

After Simon, after Hrushovski

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ABSTRACT. This is a rewriting of [Si] and [Hr]. But I took a few liberties in the setting and the notation.

E.g. I use a more compact notation when defining the patterns language.

I work with realizations of types over M instead of types themselves, and I use the notation $\mathcal{N}$ for a suitable set of realizations of the types over M .

I restrict the formulas used to define the pattern language because I would like to accommodate the example in Section 5 – this only partially succeeds.

I replace the formulas $\alpha(z)$ with types (and coheirs with strong coheirs) as presented in appendix A of [Hr].

1. Patterns

In the following $\mathcal{U}$ is some monster model of a complete theory T of language L . For simplicity L is assumed to be one-sorted.

We define a new structure $\mathcal{N}$ whose domain is a subset of $\mathcal{U}^x$, where x is some fixed tuple of variables (as a matter of fact, we could safely assume $|x| = 1$). The language of $\mathcal{N}$, which is denoted by L^p , is defined below. First we need some notation.

Let $z = \langle z_i : i < \lambda \rangle$ be a tuple of (so-called) parameter variables. We will work with a fixed set of formulas $L_z \subseteq \Delta(x; z) \subseteq L_{x,z}$. Let V be some large set of variables that are copies of x . Finally, let Δ be the set of Boolean combinations of formulas in $\bigcup_{x' \in V} \Delta(x'; z)$. The choice of Δ is justified in Remark 1.

Let $A \subseteq \mathcal{U}^x$ and $B \subseteq \mathcal{U}^z$. We write A, Δ for the set of formulas $\varphi(\bar{a}; z)$ for $\varphi(\bar{x}; z) \in \Delta$ and $\bar{a} \in A^{\bar{x}}$, and we write Δ, B for the set of formulas $\varphi(\bar{x}; b)$ for $\varphi(\bar{x}; z) \in \Delta$ and $b \in B^z$.

Let $\bar{a}, \bar{a}' \in (\mathcal{U}^x)^n$ and $b, b' \in \mathcal{U}^z$. If $\varphi(\bar{a}; b) \leftrightarrow \varphi(\bar{a}'; b)$ holds for every $\varphi(\bar{x}; z) \in \Delta$ and $b \in B^z$, we write $\bar{a} \equiv_{\Delta, B} \bar{a}'$. We define $b \equiv_{A, \Delta} b'$ similarly.

In this notes, by a **finitary type** we mean a parameter-free type where only finitely many variables (among those displayed) occur. For every formula $\varphi(x_0, \dots, x_{n-1}; z) \in \Delta$ and

every finitary type $\alpha(z) \subseteq L$ the language L^P contains the n -ary relation $r_{\varphi;\alpha}$. We abbreviate $r_{\varphi;\top}$ with r_φ . Indeed, the reader is invited to read $\alpha(z)$ as $\top$ until Section 6, where it plays an essential role.

Below the symbol L^P will also denote the set of *atomic* formulas $r_{\varphi;\alpha}(x_0, \dots, x_{n-1})$ for any $x_0, \dots, x_{n-1} \in V$. Note that equalities are *not* included in L^P .

Given a pair of models $M \leq N$, where M is ω -saturated and ω -homogeneous and N is $|M|^+$ -saturated, we define a model $\mathcal{N}$ with domain N^x . We define the interpretation of $r_{\varphi;\alpha}$ in $\mathcal{N}$. If $\bar{a} = a_0, \dots, a_{n-1} \in N^x$, we declare $r_{\varphi;\alpha}(\bar{a})$ true if

$$r_{\varphi;\alpha}(\bar{a}) \Leftrightarrow \alpha(M^z) \cap \varphi(\bar{a}; M^z) = \emptyset.$$

We say that $\varphi'(\bar{x}; z)$ is an *alphabetic variant* of the formula $\varphi(\bar{x}; z)$ if it is obtained by replacing the z -variables effectively occurring in $\varphi(\bar{x}; z)$ by other variables among those in z . Similarly, for finitary types. Clearly, if $\varphi'(\bar{x}; z)$ and $\alpha'(z)$ are alphabetic variants of $\varphi(\bar{x}; z)$ and $\alpha(z)$ respectively, then $r_{\varphi;\alpha}(\bar{x})$ and $r_{\varphi';\alpha'}(\bar{x})$ are equivalent.

The following is paramount.

1 Remark By the choice of Δ , the truth of $r_{\varphi;\alpha}(\bar{a})$ only depends on the equivalence class of $\equiv_{\Delta, M}$ of the *components* of the tuple $\bar{a}$. Explicitly, if $\bar{a} = a_0, \dots, a_{n-1}$ and $\bar{a}' = a'_0, \dots, a'_{n-1}$ are such that $a_i \equiv_{\Delta, M} a'_i$ for every $i < n$, then

$$\# \quad r_{\varphi;\alpha}(\bar{a}) \leftrightarrow r_{\varphi;\alpha}(\bar{a}').$$

By the remark, it would be natural to work in the quotient $\mathcal{N}/\equiv_{\Delta, M}$. But we prefer to keep the original domain $\mathcal{N}$ to have a common playground with L .

The set of maximally consistent set of formulas of the form $\varphi(x; b)$, where $\varphi(x; z) \in \Delta$ and $b \in M^z$ is denoted by $S_{x, \Delta}(M)$. We will often confuse it with either $\mathcal{U}^x/\equiv_{\Delta, M}$ or $\mathcal{N}/\equiv_{\Delta, M}$, with a preference for the latter.

A *p-type* is a set of formulas in $L^P(\mathcal{N})$.

Note incidentally that the relation $x_0 \equiv_{\Delta, M} x_1$ is definable by a p-type. Namely the p-type containing the formulas $r_{\vartheta}(x_0, x_1)$ for every $\vartheta(x_0, x_1; z)$ of the form $\varphi(x_0; z) \leftrightarrow \varphi(x_1; z)$.

2 Fact Formulas in L^P are closed under adjunction of dummy variables, disjunction, conjunction, and existential quantification.

Proof. The claim about dummy variables is clear, so we tacitly use it below.

Up to replacing $\varphi(\bar{x}; z)$ and $\alpha(z)$ with alphabetic variants, we can assume that the z -variables that effectively occur in $\varphi(\bar{x}; z)$ and $\alpha(z)$ are disjoint from those that occur in $\psi(\bar{x}; z)$ and $\beta(z)$. Therefore the following equivalences hold

$$\begin{aligned} r_{\varphi;\alpha}(\bar{a}) \vee r_{\psi;\beta}(\bar{a}) &\Leftrightarrow \forall z \in \alpha(M^z) \neg \varphi(\bar{a}; z) \vee \forall z \in \beta(M^z) \neg \psi(\bar{a}; z) \\ &\Leftrightarrow \forall z \in \alpha(M^z) \cap \beta(M^z) \neg \varphi(\bar{a}; z) \vee \neg \psi(\bar{a}; z) \end{aligned}$$

A similar argument applies to the conjunction of two formulas.

Finally, consider the formula $\exists y r_{\varphi;\alpha}(\bar{x}, y)$. By the definition of Δ , we can assume that φ has the form

$$\varphi(\bar{x}, y; z) = \bigvee_{i=1}^m \varphi_i(\bar{x}, y; z)$$

where $\varphi_i(\bar{x}, y; z) = \varphi'_i(\bar{x}; z) \wedge \psi_i(y; z)$ and each $\varphi'_i(\bar{x}; z)$ is a conjunction of formulas in $\bigcup_{x' \in V} L_{x'; z}$ or negation thereof. Closure under existential quantification follows from the following equivalences

$$\begin{aligned} \exists y r_{\varphi;\alpha}(\bar{a}, y) &\Leftrightarrow \exists y \forall z \left[z \in \alpha(M^z) \rightarrow \neg \bigvee_{i=1}^m \varphi_i(\bar{a}, y; z) \right] \\ &\Leftrightarrow \forall z \left[z \in \alpha(M^z) \rightarrow \neg \left(\bigvee_{i=1}^m \varphi'_i(\bar{a}; z) \wedge \forall y \psi_i(y; z) \right) \right] \\ &\Leftrightarrow r_{\vartheta;\alpha}(\bar{a}), \end{aligned}$$

where

$$\vartheta(\bar{x}; z) = \bigvee_{i=1}^m \varphi'_i(\bar{x}; z) \wedge \forall y \psi_i(y; z).$$

As $\exists y \psi_i(y; z) \in L_z \subseteq \Delta$, the above formula is in Δ as required. $\square$

3 Fact (∞ -p-saturation) Let $A \subseteq \mathcal{N}$ have arbitrary cardinality. Let $\bar{x}$ be a tuple of copies of x . Then every p-type $p(\bar{x}) \subseteq L^P(A)$ that is finitely consistent in $\mathcal{N}$ is realized in $\mathcal{N}$.

Proof. Let $A' \subseteq \mathcal{N}$ be a small set containing a representative of every $\equiv_{\Delta, M}$ equivalence class that intersects A . Replacing the elements of A that occur in $p(\bar{x})$ by elements of A' we obtain an equivalent type $p'(\bar{x}) \subseteq L^P(A')$. Note that every formula $r_{\varphi;\alpha}(\bar{x})$ there is a Δ -type $q(\bar{x}) \subseteq L(M)$ such that

$$\mathcal{N} \models r_{\varphi;\alpha}(\bar{a}) \Leftrightarrow \mathcal{N} \models q(\bar{a})$$

Therefore the saturation of $\mathcal{N}$ implies that $p'(\bar{x})$ is realized in $\mathcal{N}$. $\square$

2. Strong coheirs

We need a notion of independence that is stronger than the usual notion of coheir. The notion is adapted from that is [KNP]. We use parameter-free types (finitary types are assumed to be parameter-free) where [KNP] uses finitely many parameters from M . We believe the difference is minor but it simplifies the exposition.

Let $p(z) \subseteq L(N)$. We say that $p(z) \subseteq L(N)$ is **strongly finitely satisfiable** in M if, for every finitary type $\alpha(z) \subseteq p$, and every formula $\varphi(z) \in p$, the type $\alpha(z) \wedge \varphi(z)$ is satisfied in M . Recall that a finitary type is one which is parameter-free and with finitely many variables.

Let $b \in \mathcal{U}^z$ and $C \subseteq N$. When $\text{tp}(b/C)$ is strongly finitely satisfiable in M , we write $b \perp_M C$. Explicitly, $b \perp_M C$ holds when $\alpha(M^z) \cap \varphi(M^z) \neq \emptyset$ for every $\varphi(z) \in L(C)$ and every finitary $\alpha(z)$ such that $b \models \alpha(z) \wedge \varphi(z)$.

4 Fact Every type $p(z) \subseteq L(N)$ that is strongly finitely satisfiable in M has an extension to a complete type $q(z) \in S(N)$ that is strongly finitely satisfiable in M .

Proof. Let $q(z) \subseteq L(N)$ be maximal among the types that contain $p(z)$ and are strongly finitely satisfiable in M . We prove that $q(z)$ is complete. Suppose not, say it does not contain $\varphi(z)$ nor $\neg\varphi(z)$. Then there is a finitary type $\alpha(z) \subseteq p$ and a formula $\psi(z) \in p$ such that neither $\alpha(z) \wedge \psi(z) \wedge \varphi(z)$ nor $\alpha(z) \wedge \psi(z) \wedge \neg\varphi(z)$. This contradicts the strong finite satisfiability of $q(z)$. $\square$

Let $\hat{m}$ be an enumeration of M and assume that $|z| = |\hat{m}|$. We define

$$\mathcal{B} = \{b \in \mathcal{U}^z : \hat{m} \equiv b \perp_M N\},$$

where $\equiv$ is the elementary equivalence in L (however, as $L_z \subseteq \Delta$, we could replace it with $\equiv_{\emptyset, \Delta}$). The N, Δ -topology on $\mathcal{B}$ is the one generated by the closed sets $\{b \in \mathcal{B} : \varphi(\bar{a}; b)\}$ for $\varphi(\bar{x}; z) \in \Delta$ and $\bar{a} \in N^{\bar{x}}$. The N, Δ -topology on $\mathcal{B}/\equiv_{N, \Delta}$ is the quotient of the topology on $\mathcal{B}$.

3. Morphism

A p-morphism is a map $f : \mathcal{N} \rightarrow \mathcal{N}$ such that for every $r_{\varphi; \alpha}(\bar{x}) \in L^p$ and every $\bar{a} \in (\text{dom } f)^{\bar{x}}$

$$\# \quad \mathcal{N} \models r_{\varphi; \alpha}(\bar{a}) \Rightarrow \mathcal{N} \models r_{\varphi; \alpha}(f\bar{a}).$$

As $\equiv_{\Delta, M}$ is definable by a p-type, p-morphism induce well-defined maps on the quotient $\mathcal{N}/\equiv_{\Delta, M}$. We say that two p-morphism f and g are **equivalent**, and write $f \sim g$, if they induce the same map on $\mathcal{N}/\equiv_{\Delta, M}$. We often confuse equivalent p-morphism and we use the same symbol f to denote the map induced on the quotient.

An **endomorphism** is a total morphism. We write $\text{End}(\mathcal{N})$ for the set of p-endomorphisms of $\mathcal{N}$.

5 Definition We endow $\text{End}(\mathcal{N})$ with the topology that has as basic closed sets those of the form

$$\left\{ f \in \text{End}(\mathcal{N}) : \mathcal{N} \models r_{\varphi; \alpha}(fa_1, \dots, fa_n, c_1, \dots, c_m) \right\}$$

for $\varphi(x; z) \in \Delta$ and $a_1, \dots, a_n, c_1, \dots, c_m \in N^x$.

Note that, by Remark 1, if $f \sim g$ then f and g belong to the same basic closed sets, therefore this topologies naturally induces a topology on $\text{End}(\mathcal{N})/\sim$.

6 Fact With the above topology $\text{End}(\mathcal{N})$ is T_1 (i.e. singletons are closed) and compact. Composition of p-morphism, which we denote by $*$, is a left continuous operation.

Proof. We prove that singletons are closed. In fact

$$\{f\} = \bigcap_{a \in N^x} \{g \in \text{End}(\mathcal{N}) : a \equiv_{\Delta, M} g a\}$$

therefore, as $\equiv_{\Delta, M}$ is defined by a p-type, $\{f\}$ is the intersection of closed sets.

We prove compactness. Assume the following intersection is finitely consistent

$$\#. \quad \bigcap_{i < \lambda} \{f \in \text{End}(\mathcal{N}) : \mathcal{N} \models r_{\varphi_i; \alpha_i}(f \bar{a}, \bar{c})\}.$$

then the p-type

$$\bar{x} \equiv_{\Delta, M} \bar{a} \cup \{r_{\varphi_i; \alpha_i}(\bar{x}, \bar{c}) : i < \lambda\}$$

is finitely consistent in $\mathcal{N}$. By ∞ -p-saturation, it is realized by some $\bar{a}'$. By Fact 7 there is a p-endomorphism $\mathcal{N}$ that maps $\bar{a}$ to $\bar{a}'$. This proves that the set in # is non empty.

We prove left continuity of composition. That is for any given $f \in \text{End}(\mathcal{N})$ we prove that the function $\tau_f : \text{End}(\mathcal{N}) \rightarrow \text{End}(\mathcal{N})$ that maps $g \mapsto g * f$ is continuous. Pick any basic closed set $\{h : r_{\varphi; \alpha}(h \bar{a}, \bar{c})\}$. Then

$$\tau_f^{-1}[\{h : r_{\varphi; \alpha}(h \bar{a}, \bar{c})\}] = \{g : r_{\varphi; \alpha}(g(f \bar{a}), \bar{c})\}$$

is closed, as required. $\square$

If the converse of the implication in # also holds, f is called a **partial p-isomorphism**. For the back-and-forth argument in the following proof we need to take inverse of partial p-isomorphisms. Strictly speaking partial p-isomorphism need not be invertible but for our puposes it suffices to have a p-morphism g such that $f \circ g$ and $g \circ f$ are equivalent to the identity on $\text{range } f$, respectively $\text{dom } f$.

7 Fact Every p-morphism extends to a p-morphism defined on the whole of $\mathcal{N}$. Also, every p-isomorphism extends to a p-automorphism of $\mathcal{N}$.

Proof. Let $f : \mathcal{N} \rightarrow \mathcal{N}$ be a p-morphism. Let $\bar{a}$ be an enumeration of $\text{dom } f$ and let $b \in N^x$. Let $p(\bar{x}) = \text{p-tp}(\bar{a})$ and let $q(\bar{x}, y) = \text{p-tp}(\bar{a}, b)$. By ∞ -p-saturation $\exists y q(\bar{x}, y)$ is equivalent to the type containing the existential closure of the formulas in $q(\bar{x}, y)$. By Fact 2, the latter is equivalent to a p-type, which is necessarily coincides with $p(\bar{x})$. Then $q(f \bar{a}, y)$ is consistent hence by ∞ -p-saturation it is realized by some $c \in N^x$. Then $f \cup \{\langle b, c \rangle\}$ is a p-morphism defined in b that extends f .

The second claim is obtained by back-and-forth. $\square$

4. Duality

Recall that $\hat{m}$ is an enumeration of M and $|z| = |\hat{m}|$. For convenient notation, x is assumed to be of infinite length – then, up to equivalence, all formulas in $L_z(\mathcal{U})$ have the form $\varphi(a; z)$ for some $\varphi(x; z) \in L$ and $a \in \mathcal{U}^x$.

8 Remark We stress a simple fact which is used in the proof below. Recall that $\varphi'(a; z)$ and $\alpha'(z)$ are alphabetic variants of $\varphi(a; z)$ and $\alpha(z)$ if they are obtained by replacing the variables effectively occurring in $\varphi(a; z)$ and $\alpha(z)$ by other variables among those in z . The following are clearly equivalent

1. $\alpha(M^z) \cap \varphi(a; M^z) = \emptyset$
2. $\hat{m} \not\models \alpha'(z) \wedge \varphi'(a; z)$ for every alphabetic variants of $\varphi(a; z)$ and $\alpha(z)$.

9 Fact There is a continuous bijection $\mathcal{B}/\equiv_{N, \Delta} \rightarrow \text{End}(\mathcal{N})/\sim$.

Proof. In 1 below, we define a map $b \mapsto f_b$ from $\mathcal{B}$ into $\text{End}(\mathcal{N})$. In 2, we define a map $f \mapsto b_f$ from $\text{End}(\mathcal{N})$ into $\mathcal{B}$. The reader may check that these maps induce maps between $\mathcal{B}/\equiv_{N, \Delta}$ and $\text{End}(\mathcal{N})/\sim$ that are each other's inverse. In 3 we prove the continuity of the first map.

1. Let $b \in \mathcal{B}$ be given. We define

$$p(x, y, z) = \{ \varphi(y; z) \rightarrow \varphi(x; \hat{m}) : \varphi(x; z) \in \Delta \}.$$

We claim that if $a \equiv_{\Delta, M} a'$ then $p(x, a, b) = p(x, a', b)$. In fact, $\varphi(a; b) \leftrightarrow \varphi(a'; b)$ would contradict $b \perp_M N$. Also, note that $p(x, a, b)$ is consistent, otherwise $\neg \exists x \bigwedge_i \varphi_i(x; \hat{m})$, for some finitely many formulas $\varphi_i(x; z) \in \Delta$ such that $\varphi_i(a; b)$. But this would contradict $b \equiv \hat{m}$.

Therefore it is possible to define $f_b(a)$ to be some/any realization of $p(x, a, b)$. We prove that f_b is a p-morphism.

Let $\psi(x_1, \dots, x_n; z) \in \Delta$, $\alpha(z) \subseteq L$, and $a_1, \dots, a_n \in \mathcal{U}^x$ be such that $\neg r_{\psi; \alpha}(f_b a_1, \dots, f_b a_n)$. Then $\alpha(M^z) \cap \psi(f_b a_1, \dots, f_b a_n; M^z) \neq \emptyset$. Pick some alphabetic variants such $\alpha'(\hat{m})$ and $\psi'(f_b a_1, \dots, f_b a_n; \hat{m})$. From the definition of f_b we obtain $\psi'(a_1, \dots, a_n; b)$ and, as $b \equiv \hat{m}$, we also have $\alpha'(b)$. Finally, as $b \perp_M N$, we conclude that $\alpha'(M^z) \cap \psi'(a_1, \dots, a_n; M^z) \neq \emptyset$, i.e. that $\neg r_{\psi; \alpha}(a_1, \dots, a_n)$.

2. Let $f : \mathcal{N} \rightarrow \mathcal{N}$ be an p-epimorphism. Let $p(z)$ be the type containing the formulas $\varphi(a; z)$ for $\varphi(x; z) \in \Delta$ and $a \in N^x$ such that $\varphi(fa; \hat{m})$. Clearly $p(z) \vdash \hat{m} \equiv z$ and is complete over N, Δ . We prove that $p(z)$ is strongly finitely satisfied in M , then b_f is some/any realization of $p(z)$.

Suppose for a contradiction $p(z)$ is not strongly finitely satisfied in M . Then $\alpha(M^z) \cap \psi(a_1, \dots, a_n; M^z) = \emptyset$, where $\alpha(z) \subseteq p$ is a finitary type and

$$\psi(x_1, \dots, x_n; z) = \bigwedge_{i=1}^n \varphi_i(x_i; z)$$

for some $\varphi_i(x; z) \in L$ and $a_i \in N^x$ such that $\varphi_i(fa_i; \hat{m})$ for every i . Then $r_{\psi; \alpha}(a_1, \dots, a_n)$ and $\neg r_{\psi; \alpha}(f a_1, \dots, f a_n)$. This is not possible because f is a p-morphism.

3. We prove that the map defined in 1 is continuous. Pick any basic closed set in $\text{End}(\mathcal{N})$, say $\{h : r(h\bar{a}, \bar{c})\}$. We claim that the set $\{b : r_{\psi; \alpha}(f_b \bar{a}, \bar{c})\}$ is closed. Indeed, by the definition of f_b in 1,

$$\{b : r_{\psi;\alpha}(f_b \bar{a}, \bar{c})\} = \left\{b : b \models \exists \bar{x} \left[\bigwedge_{i=1}^n p(x_i, a_i, z) \wedge r_{\psi;\alpha}(\bar{x}, \bar{c}) \right] \right\}$$

is closed in $\mathcal{B}$ because $r_{\psi;\alpha}(\bar{x}, \bar{c})$ is definable by a N, Δ -type. $\square$

5. Semigroups

Assume $\mathcal{U}$ is a unital semigroup (a monoid) w.r.t. some definable operation which we denote by $\cdot$. The unit is denoted by 1. In this section we could work with $|x| = |z| = 1$ but, for uniformity with the previous section we take $|x| = |z| = \omega$. We write $z \cdot x$ for coordinatewise multiplication. In this section $\Delta(x; z)$ contains the formulas $\varphi(z \cdot x)$ for all $\varphi(x) \in L$.

10 Fact Assume that $a \cdot M = M$ for all $a \in M$, which holds in particular when $\mathcal{U}$ is a group. For $b \in \mathcal{B}$, let $g_b : \mathcal{N} \rightarrow \mathcal{N}$ be the function that maps $a \mapsto b \cdot a$. Then $b \mapsto g_b$ induces a bijection between $\mathcal{B}/\equiv_{N, \Delta}$ and $\text{End}(\mathcal{N})/\sim$.

Proof. In 1 below we prove that g_b is a p-morphism. In 2 we define a map $g \mapsto b_g$ from $\text{End}(\mathcal{N})$ into $\mathcal{B}$. The reader may check that these maps induce maps between $\mathcal{B}/\equiv_{N, \Delta}$ and $\text{End}(\mathcal{N})/\sim$ that are each other's inverse. In 3 we prove the continuity of the map $b \mapsto g_b$.

1. Assume $r_{\psi;\alpha}(a_1, \dots, a_n)$. We prove $r_{\psi;\alpha}(g_b a_1, \dots, g_b a_n)$. Suppose not, then $\alpha(M^z) \cap \psi(M^z \cdot b \cdot a_1, \dots; M^z \cdot b \cdot a_n) \neq \emptyset$. From $b \perp_M N$, we obtain $\alpha(M^z) \cap \psi(M^z \cdot a_1, \dots; M^z \cdot a_n) \neq \emptyset$. This contradicts $r_{\psi;\alpha}(a_1, \dots, a_n)$.

2. Let $g : \mathcal{N} \rightarrow \mathcal{N}$ be a p-morphism. Let $p(y)$ be the type containing $\varphi(\hat{m} \cdot y \cdot a)$ for all $\varphi(x) \in L$ and $a \in N^x$ such that $\varphi(\hat{m} \cdot ga)$. We prove that $p(y)$ is strongly finitely satisfied in M . If for a contradiction $\alpha(M^z) \cap \psi(a_1, \dots, a_n; \hat{m} \cdot M^z) = \emptyset$, where

$$\psi(x_1, \dots, x_n; z) = \bigwedge_{i=1}^n \varphi_i(z \cdot x_i)$$

for some $\varphi_i(x) \in p$ and $a_i \in N^x$ such that $\varphi_i(\hat{m} \cdot ga_i)$ for every i . As $\hat{m} \cdot M^z = M^z$, we have that $r_{\psi}(a_1, \dots, a_n)$ and $\neg r_{\psi}(ga_1, \dots, ga_n)$. This is not possible because g is a p-morphism. Then any b_g that realizes $p(z)$ is such that $g \sim b_g$.

3. We prove that the map defined in 1 is continuous. Pick any basic closed set in $\text{End}(\mathcal{N})$, say $\{h : r(h\bar{a}, \bar{c})\}$ where $\bar{a} = a_1, \dots, a_n$ and $\bar{c} = c_1, \dots, c_m$. We claim that the set $\{b : r(g_b \bar{a}, \bar{c})\}$ is closed. Indeed, by the definition of g_b in 1,

$$\{b : r(g_b \bar{a}, \bar{c})\} = \{b : r(b \cdot a_1, \dots, b \cdot a_n, \bar{c})\}$$

is closed in $\mathcal{B}$ because $r(w \cdot a_1, \dots, w \cdot a_n, \bar{c})$ is definable by a N, Δ -type $p(w, \bar{a}, \bar{c})$. $\square$

6. Ellis's semigroup

We write $\text{Aut}(N/\{M\})$ for the group of automorphisms of N that fix M setwise. In this section we show that $\text{End}(\mathcal{N})$, endowed with a topology that is finer than that introduced

above, is homeomorphic to the Ellis semigroup associated to the action $\text{Aut}(N/\{M\})$ on N^x . Note that this is essentially equivalent to the action of $\text{Aut}(M)$ on $S_{x,\Delta}(M)$ – we chose the former based on personal preference.

We recall the definition of this semigroup, which we denote by $\text{El}_x(N, M)$. On $\mathcal{N}$ we assume the topology generated by the closed sets of the form $\varphi(N^x; b)$ for $\varphi(x; z) \in \Delta$ and $b \in M^z$. This we call the **logic topology**. We identify $\text{Aut}(N/\{M\})$ with a subset of $(N^x)^{N^x}$, the set of all maps from N^x into itself, which we endow with the product topology. The Ellis semigroup is the closure of $\text{Aut}(N/\{M\})$ in the product topology. The following theorem proves that $\text{El}_x(N, M)$ coincides with $\text{End}(\mathcal{N})$.

11 Theorem Assume M is λ -homogeneous (where $\lambda = |z|$). Then $\text{End}(\mathcal{N})$ is a closed subset of $(N^x)^{N^x}$, and $\text{Aut}(N/\{M\})$ is dense subset of it.

Note that the topology that $\text{End}(\mathcal{N})$ inherits from $(N^x)^{N^x}$ is finer than the topology introduced in Definition 5.

Proof. It is clear that $\text{End}(\mathcal{N})$ is a closed set containing $\text{Aut}(N/\{M\})$. Let $f \in \text{End}(\mathcal{N})$ and $a_1, \dots, a_n \in N^x$ be given. Let $\varphi_i(x; z) \in \Delta$ and $b \in M^z$ be such that $f a_i \models \varphi_i(x; b)$. We need to prove that there is $g \in \text{Aut}(N/\{M\})$ such that $g a_i \models \varphi_i(x; b)$. Let $\alpha(z) = \text{tp}(b)$. Let

$$\psi(x_1, \dots, x_n; z) = \neg \bigwedge_{i=1}^n \varphi_i(x_i; z)$$

As $\neg r_{\psi; \alpha}(f a_1, \dots, f a_n)$ and f is a p-morphism, $\neg r_{\psi; \alpha}(a_1, \dots, a_n)$. Then there is $c \models \alpha(z)$ such that $\varphi_i(a_i; c)$ for every i . As M is λ -homogeneous there is a $g \in \text{Aut}(N/\{M\})$ such that $g c = b$, then $\varphi_i(g a_i; b)$ as required. $\square$

7. Cores

A **p-retraction** of $\mathcal{N}$ onto $\mathcal{A}$ is a total p-morphism $f : \mathcal{N} \rightarrow \mathcal{A}$ that is (equivalent to) the identity on $\mathcal{A}$. Retractions are exactly the idempotent elements of $\text{End}(\mathcal{N})$. In this section we characterize the minimal idempotents of $\text{End}(\mathcal{N})$, that is, the p-morphisms $u \in \text{End}(\mathcal{N})$ such that $\text{End}(\mathcal{N}) * u$ is a minimal left ideal. These are the retractions onto substructures of $\mathcal{N}$ named **cores**.

A set $A \subseteq \mathcal{N}$ is **p-maximal** if it satisfies the following (obviously) equivalent conditions for every $\bar{a} \in A^{<\omega}$ and every p-morphism f defined on $\bar{a}$:

- a. $r_\varphi(\bar{a}) \Leftrightarrow r_\varphi(f\bar{a})$ for every $r_\varphi \in L^P$
- b. $\text{p-tp}(\bar{a}) = \text{p-tp}(f\bar{a})$.

12 Fact The following are equivalent

1. $A \subseteq \mathcal{N}$ is p-maximal
2. $p(\bar{x}) = \text{p-tp}(\bar{a})$ is a maximally consistent p-type for every $\bar{a} \in A^{\bar{x}}$ and every finite $\bar{x}$
3. $p(\bar{y}) = \text{p-tp}(\bar{c})$ is a maximally consistent p-type for $\bar{c}$ any enumeration of A
4. every p-morphism defined on A is a p-isomorphism between A and its image.

Proof. $1 \Rightarrow 2$. Negate 2. Suppose $\text{p-tp}(\bar{a})$ is not maximal for some finite $\bar{a} \in A^{\bar{x}}$. Pick some $\bar{a}' \in A^{\bar{x}}$ such that $\text{p-tp}(\bar{a}) \subset \text{p-tp}(\bar{a}')$. Then $f = \{\langle \bar{a}, \bar{a}' \rangle\}$ is a p-morphism that contradicts the p-maximality of A .

$2 \Rightarrow 3$. If $\bar{c}$ is an enumeration of A that witnesses the failure of 3, then some finite subtuple of $\bar{c}$ witnesses the failure of 2.

$3 \Rightarrow 4$. Negate 4. Let f be a p-morphism defined on A that is not a p-isomorphism. Let $\bar{a} \in A^{\bar{x}}$ be such that $\text{p-tp}(\bar{a}) \setminus \text{p-tp}(f\bar{a}) \neq \emptyset$. To contradict 3, it suffices to extend f to a p-morphism defined on A . This is possible by ∞ -p-homogeneity.

$4 \Rightarrow 1$. Clear. □

13 Example Assume that $a \in N^x$ is such that $\varphi(a; M^z)$ is definable over $\text{acl}^{\text{eq}} \emptyset$ for every $\varphi(x; z) \in L$. Then the orbit of a under $\text{Aut}(\mathcal{U}/\{M\})$ is p-maximal.

Proof. First consider the case when $\varphi(a; M^z)$ is definable over $\emptyset$. We prove that $\{a\}$ is p-maximal.

Let $\varphi'(z) \in L$ be such that $\varphi(a; M^z) = \varphi'(M^z)$. Then, if we set $\varphi''(x; z) = \varphi(x; z) \leftrightarrow \varphi'(z)$, the formula $r_{\varphi''}(a)$ holds.

If f is a p-morphism then $r_{\varphi''}(fa)$ also holds. Then $a \equiv_{\Delta, M} fa$ follows. This shows that p-morphisms are trivial, i.e. equivalent to the identity, and therefore p-isomorphisms.

Now assume that $\varphi(a; M^z)$ is definable over $\text{acl}^{\text{eq}} \emptyset$. □

A set $\mathcal{J} \subseteq \mathcal{N}$ is called a **core** if it is p-maximal and, among the p-maximal sets, it is maximal w.r.t. inclusion.

14 Fact Let $\mathcal{J}$ be a core. Let $\mathcal{A}$ be a p-maximal set. Then every p-morphism $f : \mathcal{A} \rightarrow \mathcal{J}$ extends to a p-isomorphism between $\mathcal{A}$ and $\mathcal{J}$.

Proof. Reason as in the proof of Fact 7 with the same notation but choose $b \in \mathcal{A}$. Let f' be a p-morphism defined on $\mathcal{A}$ that extends $f \cup \{\langle b, c \rangle\}$. Then $f'[\mathcal{A}]$, i.e. the image of $\mathcal{A}$ under f' , is a p-maximal set that contains $\mathcal{J}$. By maximality $f'[\mathcal{A}]$ coincides with $\mathcal{J}$, hence $c \in \mathcal{J}$. □

By back-and-forth we obtain the following.

15 Corollary (∞ -p-homogeneity of the core) If $\mathcal{J}$ and $\mathcal{J}'$ are cores, then every p-morphism $f : \mathcal{J} \rightarrow \mathcal{J}'$ extends to a p-isomorphism between $\mathcal{J}$ and $\mathcal{J}'$.

A **p-retraction** of $\mathcal{N}$ onto $\mathcal{A}$ is a total p-morphism $f : \mathcal{N} \rightarrow \mathcal{A}$ that is equivalent to the identity on $\mathcal{A}$.

16 Theorem If $\mathcal{J}$ and $\mathcal{A}$ is a core, then there is a p-retraction of $\mathcal{N}$ onto $\mathcal{J}$.

Proof. Let $\bar{c}$ enumerate $\mathcal{N}$. Let $\bar{a}$ be a tuple such that $\text{p-tp}(\bar{a}) \supseteq \text{p-tp}(\bar{c})$ is p-maximal. Let $f = \{\langle \bar{c}, \bar{a} \rangle\}$. Then $f : \mathcal{N} \rightarrow \mathcal{A}$ is a p-morphism onto the p-maximal set $\mathcal{A} = \text{range}(\bar{a})$. Then $f \upharpoonright \mathcal{J}$ is a p-isomorphism from $\mathcal{J}$ into $\mathcal{A}$. As $\mathcal{J}$ is a core, $f \upharpoonright \mathcal{J}$ is p-isomorphism between $\mathcal{J}$ and $\mathcal{A}$. Then $f \circ (f \upharpoonright \mathcal{J})^{-1}$ is a retraction of $\mathcal{N}$ onto $\mathcal{J}$. $\square$

17 Fact The following are equivalent

1. $a \in \bigcap \{ \mathcal{J} : \mathcal{J} \text{ is a core} \}$
2. $\varphi(a; M^Z)$ is definable over $\text{acl}^{\text{eq}} \emptyset$, for every formula $\varphi(x; z) \in L$.

Proof. ??? $\square$

18 Fact $\mathcal{J}$ is compact subset of $\mathcal{N}$.

Proof. ??? $\square$

8. Esperimenti

Recall that $\hat{m}$ is an enumeration of M and $|z| = |\hat{m}|$.

The following fact is well-known. It entails that the relation a and b are at distance ≤ 1 in the Lascar graph is type-definable (this is the relation defined by 1 in the fact).

19 Fact The following are equivalent

1. the $a \equiv_{M'} b$ for some model M'
2. $a \equiv_{m'} b$ for some $m' \equiv \hat{m}$.

Proof. $2 \Rightarrow 1$. Clear.

$1 \Rightarrow 2$. Negate 2. Then $\alpha(z) \vdash \varphi(a; z) \leftrightarrow \varphi(b; z)$ for some $\varphi(x; z) \in L$ and some $\alpha(z) \in \text{tp}(\hat{m})$. As every model witnesses $\alpha(z)$, this negates 1. $\square$

Let $l_1(x, x') \subseteq L$ be the type that says that x, x' are at distance ≤ 1 in the Lascar graph.

20 Fact There is a p-type defining $l_1(x, x')$.

Proof. Consider the p-type containing the formulas $r_{\vartheta, \alpha}(x_0, x_1)$ for every $\vartheta(x_0, x_1; z)$ of the form $\varphi(x_0; z) \leftrightarrow \varphi(x_1; z)$, where $\alpha(z)$ is the type $\text{tp}(\hat{m})$ restricted to the z -variables that effectively occur in $\varphi(x; z) \in \Delta$. $\square$

21 Fact If $\neg l_1(a, b)$, then $fa \not\equiv_M fb$, for every p-morphism f .

Proof. Assume $\neg l_1(a, b)$. Then $\alpha(z) \vdash \varphi(a; z) \leftrightarrow \varphi(b; z)$ for some $\varphi(x; z) \in L$ and some $\alpha(z) \in \text{tp}(\hat{m})$. Then $r_{\vartheta, \alpha}(a, b)$, where $\vartheta(x_0, x_1; z)$ is the formula $\varphi(x_0; z) \leftrightarrow \varphi(x_1; z)$. Then $r_{\vartheta, \alpha}(fa, fb)$ also holds. As $\alpha(z)$ is satisfiable in M , we obtain $fa \not\equiv_M fb$. $\square$

22 Corollary If $f : N \rightarrow \mathcal{J}$ is a retraction, then $l_1(a, fa)$ for every $a, b \in N$.

9. Notes and references

- [Hr] Ehud Hrushovski, *Definability patterns and their symmetries*, arXiv:1911.01129.
- [KNP] Krzysztof Krupiński, Ludomir Newelski, and Pierre Simon, *Boundedness and absoluteness of some dynamical invariants in model theory*, arXiv:95100.5071.
- [Si] Pierre Simon, *Notes on Hrushovski's "Definability patterns and their symmetries"*.